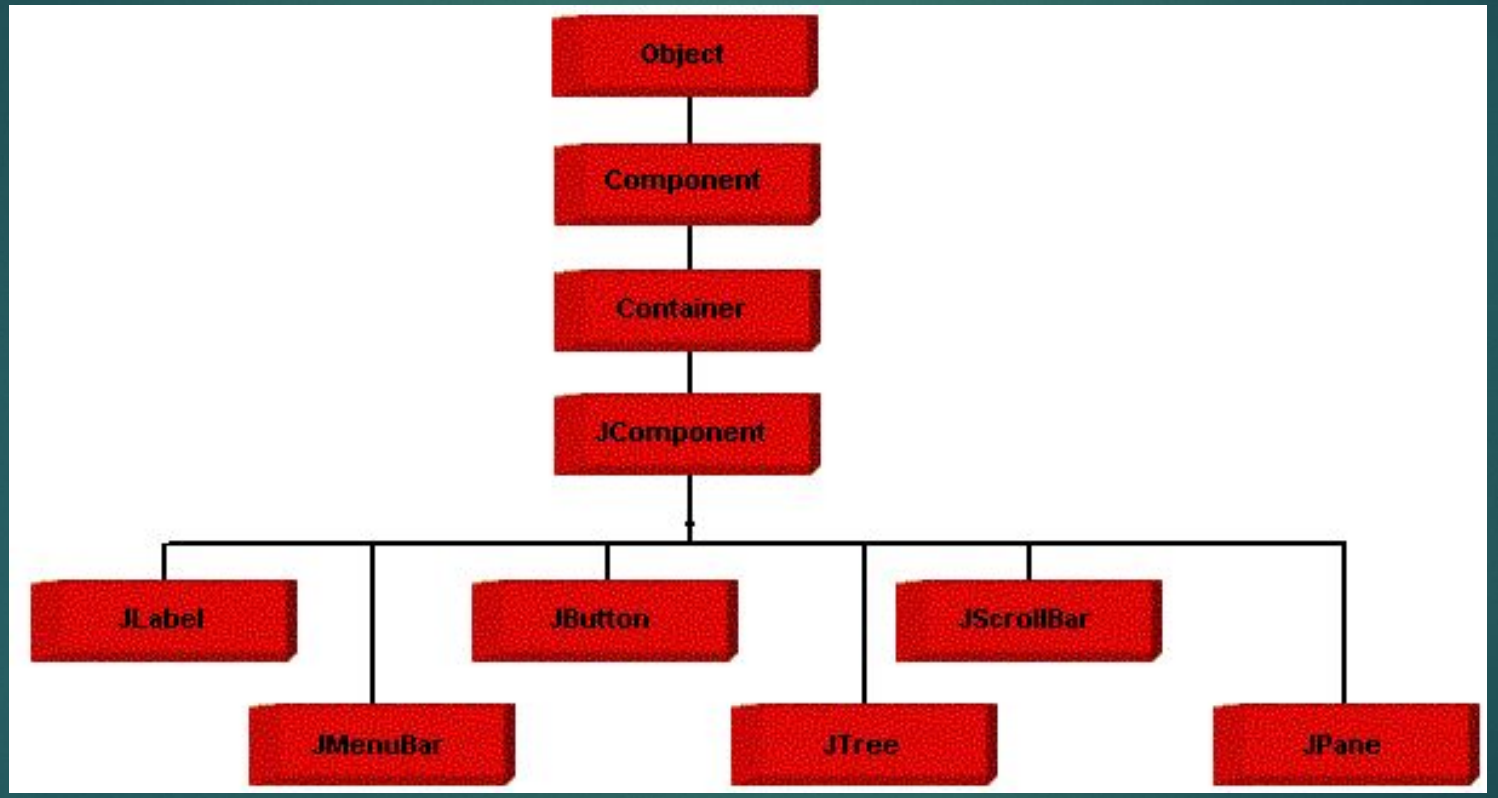
Chat Application using Sockets

Swing

- **Swing Java** consists of
  Look and feel
  Accessibility
  Java 2D
  Drag and Drop, etc
- Swing, which is an extension library to the AWT, includes new and improved components that enhance the look and functionality of GUIs.
- Swing can be used to build Standalone swing gui Apps as well as Servlets and Applets.
- It employs a model/view design architecture. Swing is more portable and more flexible than AWT.

Swing Hierarchy



JFrame

```
public class ui{

    public static void main( String args[] ) {

        JFrame win = new JFrame( "My Chat
Application" );

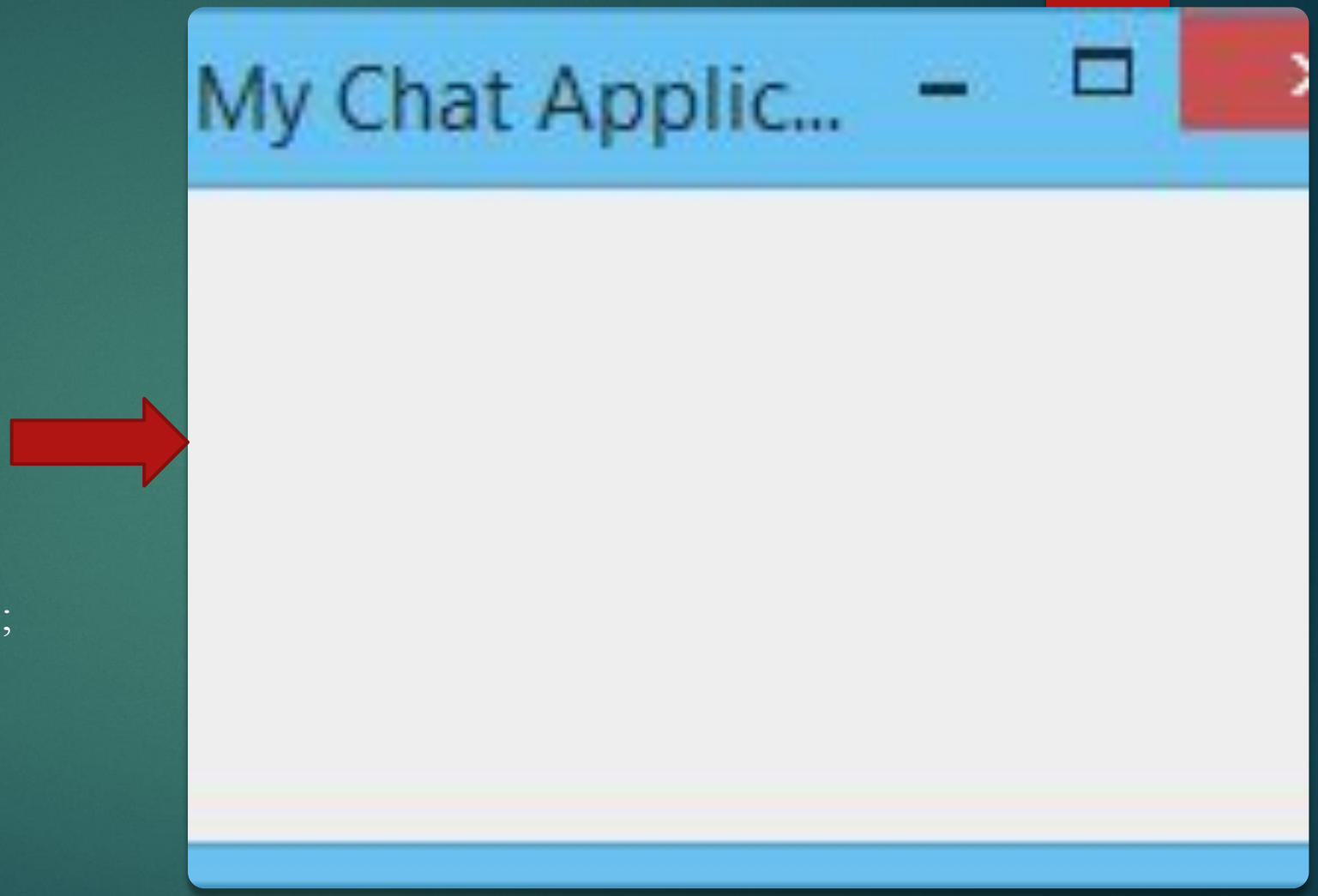
        win.setDefaultCloseOperation(
JFrame.EXIT_ON_CLOSE );

        win.setSize(new Dimension( 250, 150 ));

        win.setVisible(true);

    }

}
```



JPanel

```
public class ui {

    public static void main( String args[] ) {

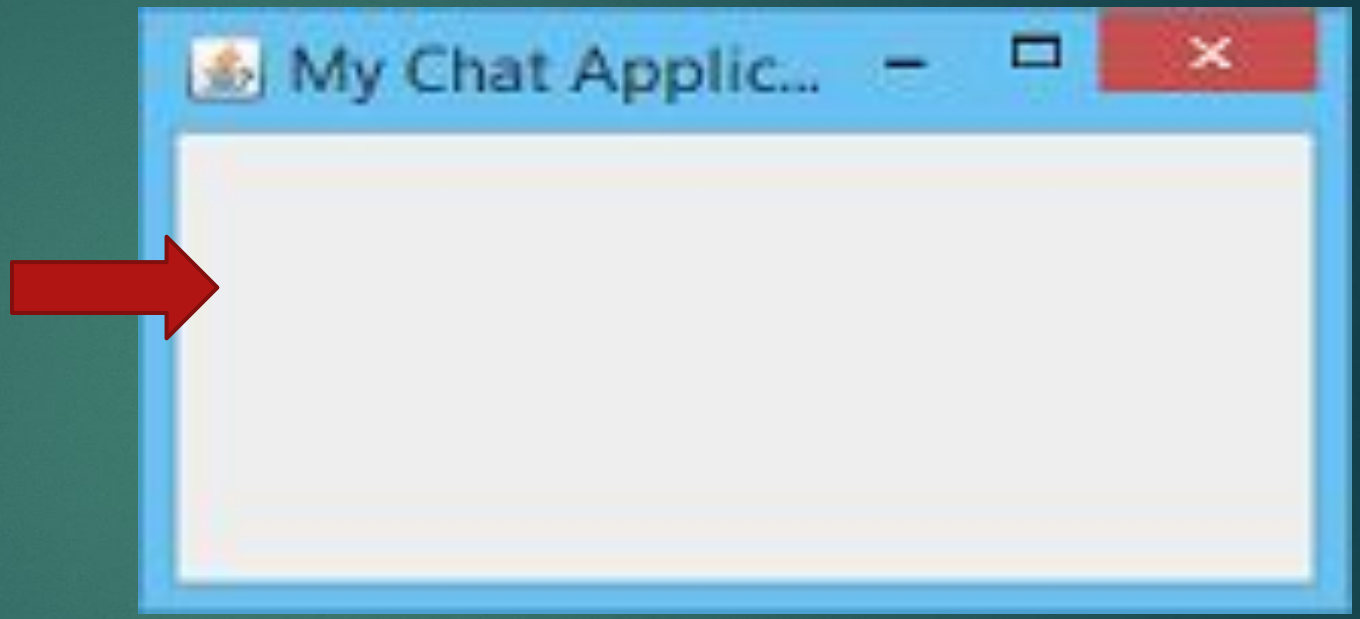
        JFrame win = new JFrame( "My Chat Application" );

        JPanel  panel = new Jpanel();

        win.add(panel):

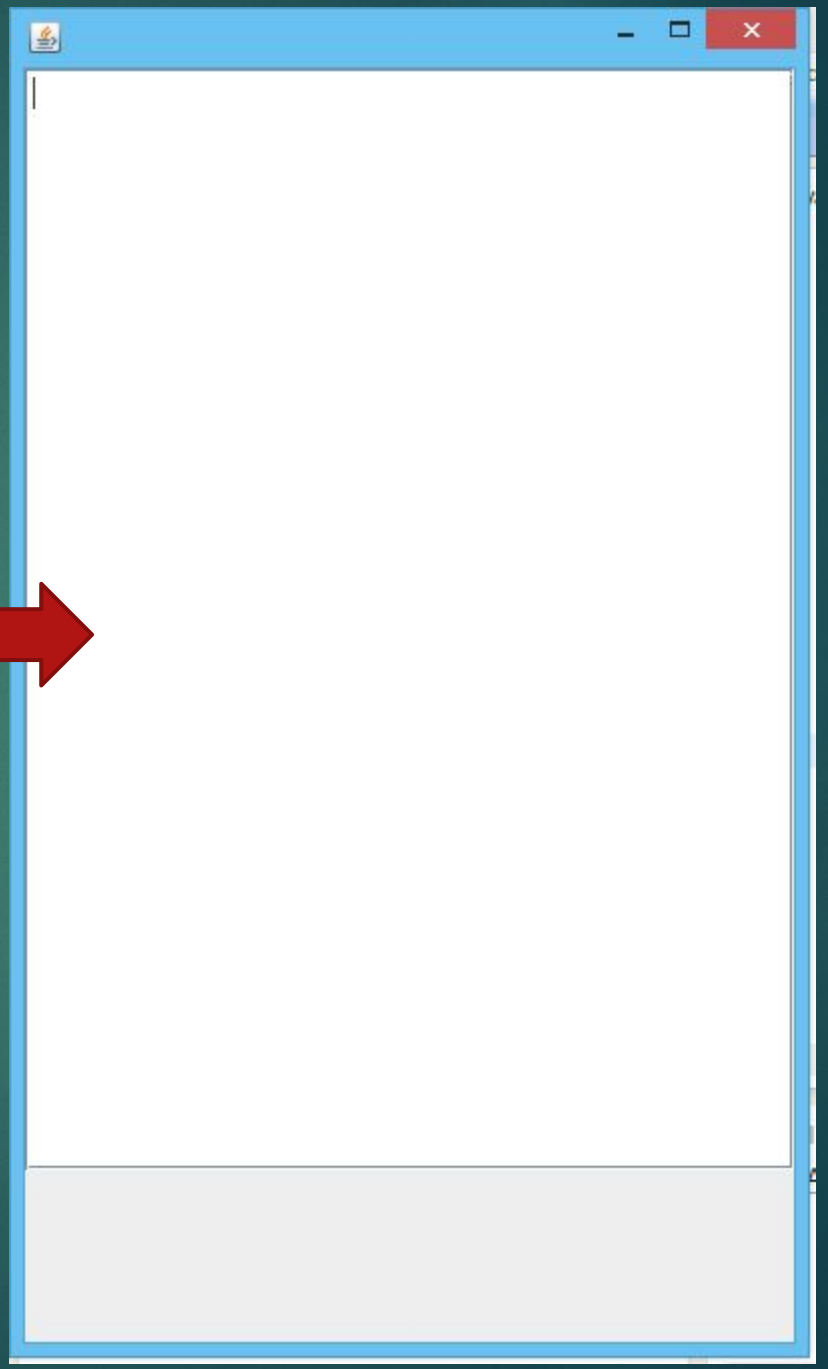
}

}
```



JTextPane

```
public static void main(String args[])
{
        JTextPane msg=new JTextPane();
        msg_view.setPreferredSize(new
Dimension(40,400));
JScrollPane sp=new JScrollPane(msg_view);
panel.add(sp,BorderLayout.NORTH);
frame.add(panel,BorderLayout.NORTH);
}
```



Adding Styles to text

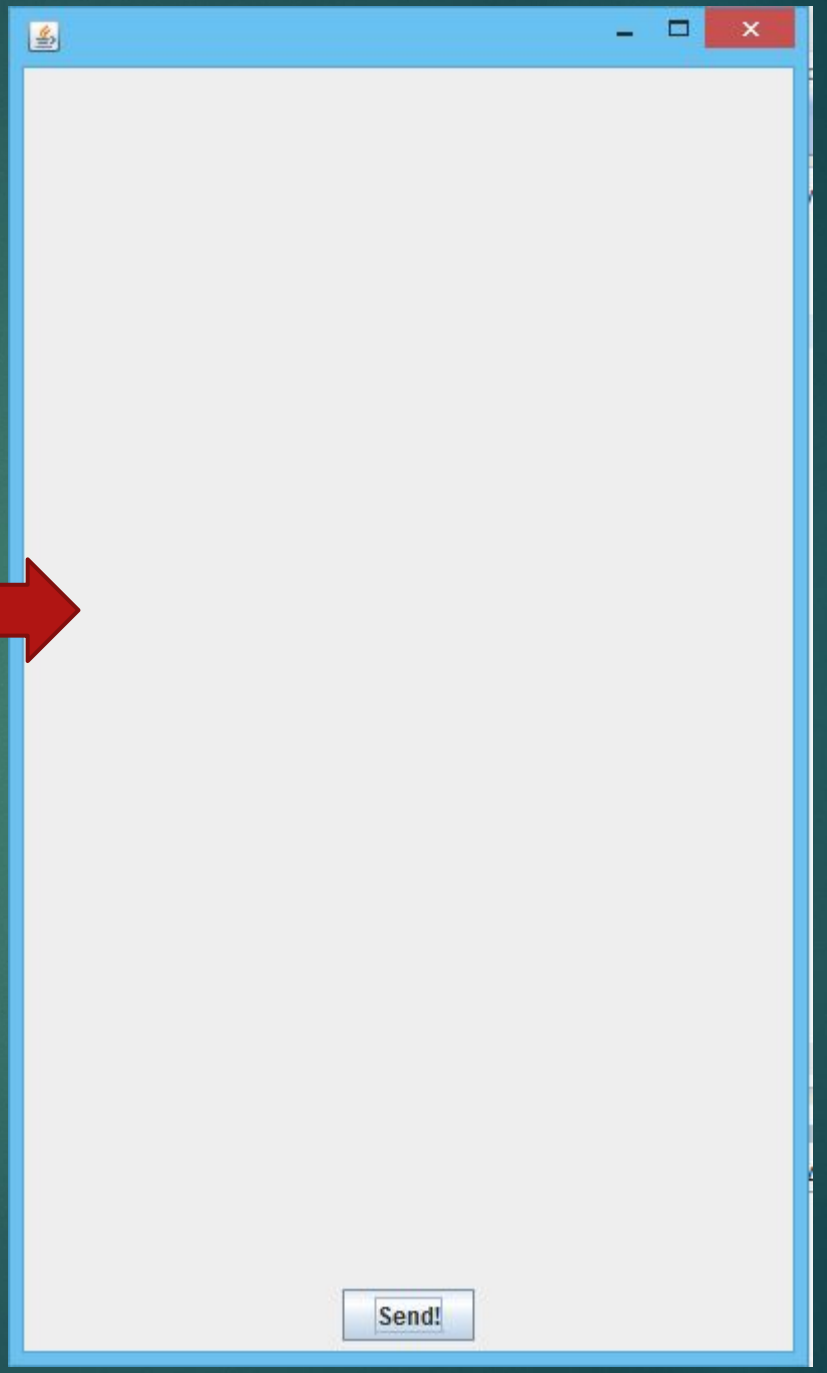
To differentiate user1 and user2 text from each other.

Let us give colors to user’s text separately.

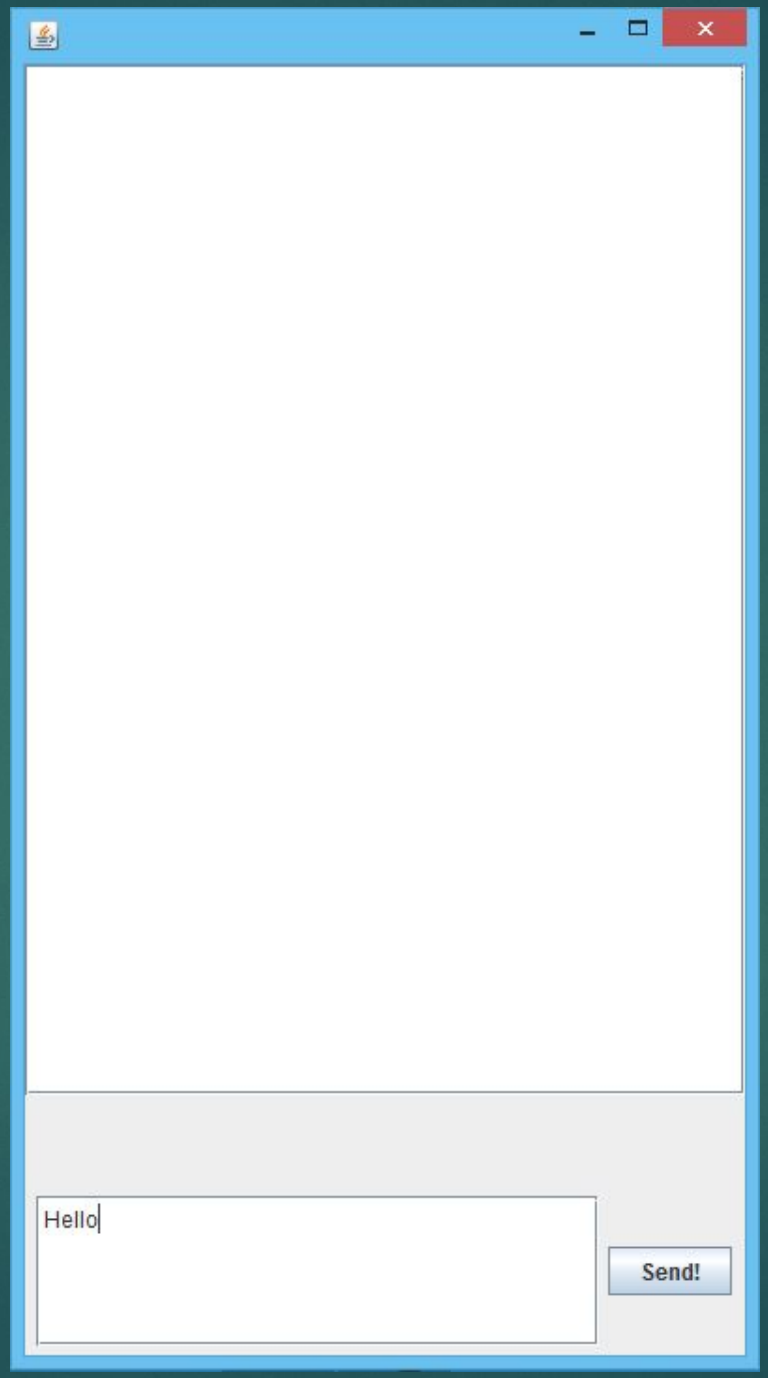
```
Style style = msg_view.addStyle("I'm a Style", null);
StyleConstants.setForeground(style, Color.blue);
```

JButton

```
public static void main(String args[])
{
        send=new JButton("Send!");
        panel2.add(send);
        frame.add(panel2,BorderLayout.SOUTH);
}
```



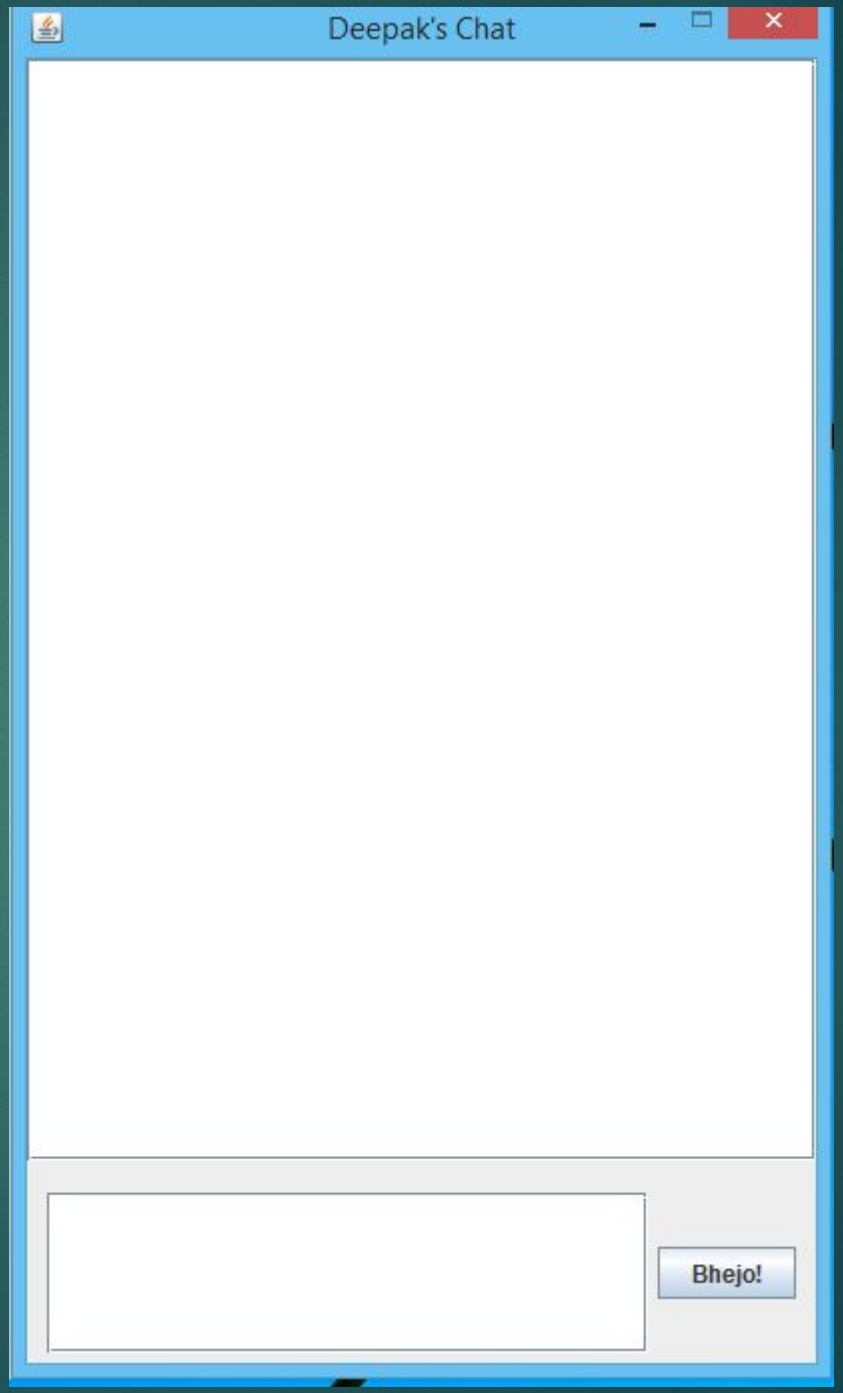
Chat Window

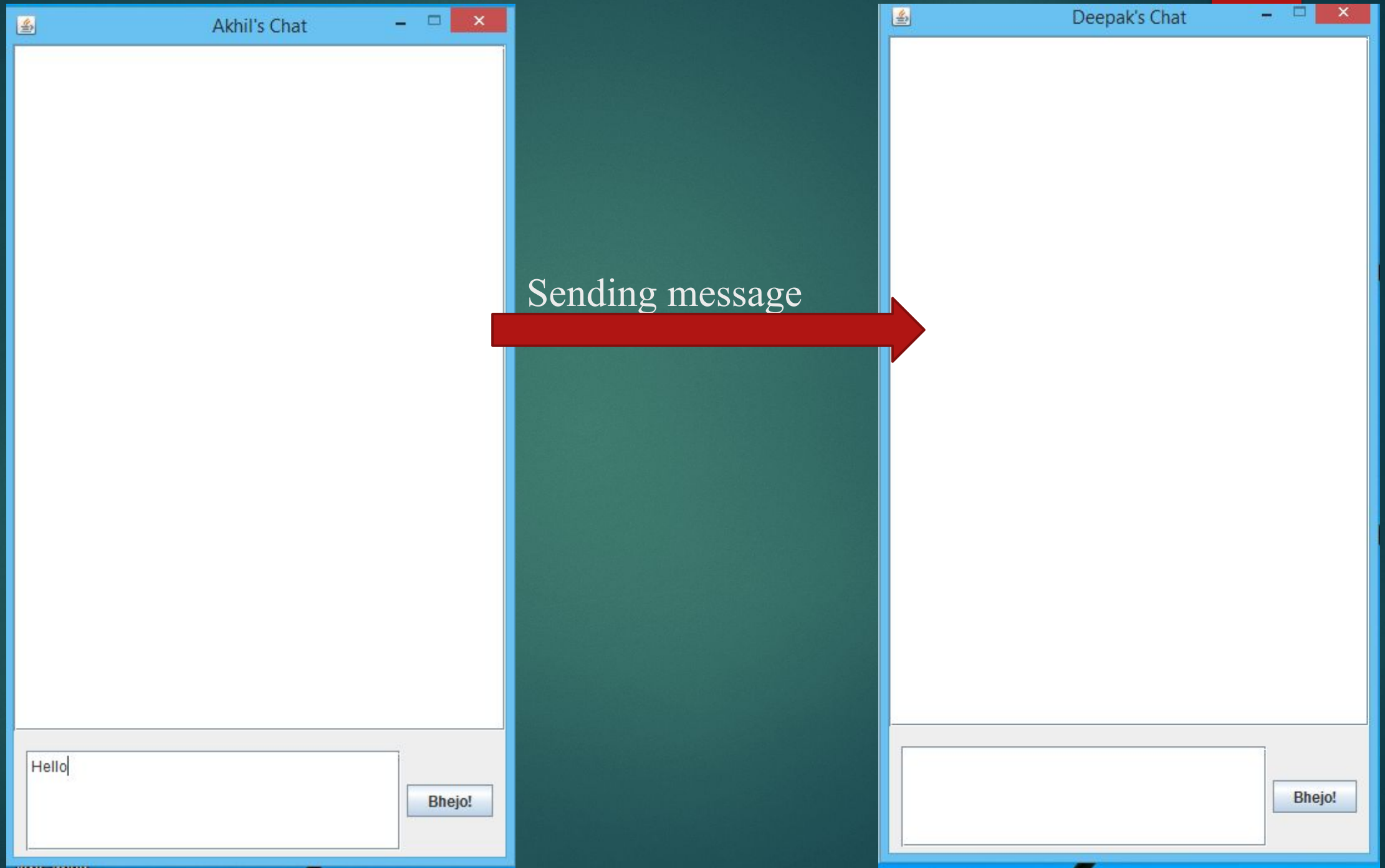


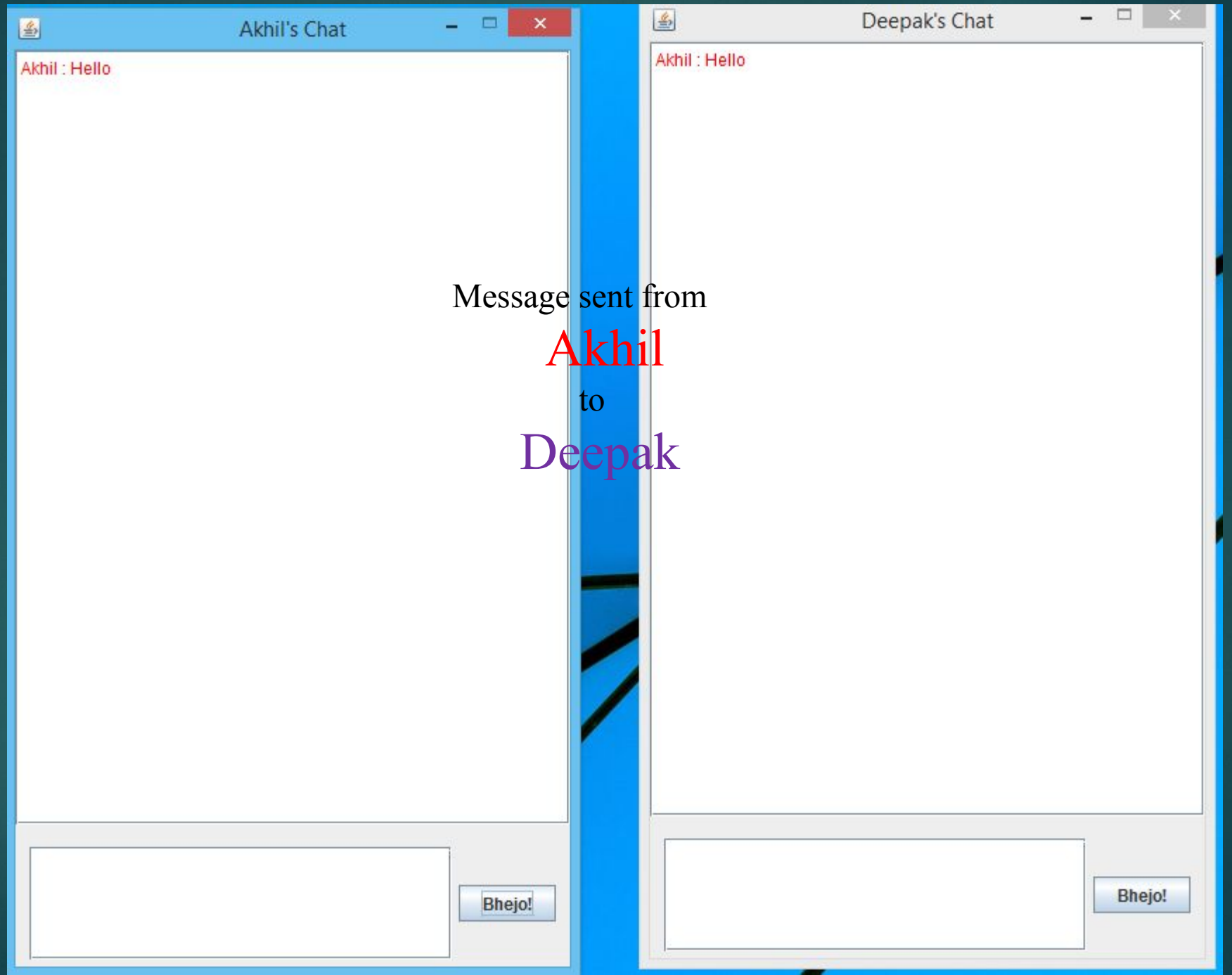
Sending and Receiving messages

- Both users must be able to send and receive messages.
- Receiving must be done continuously
- Sending must be done as an action of button “send!”click


Akhil's Chat
Bhejo!


Deepak's Chat
Bhejo!


Akhil's Chat
Hello
Bhejo!
Sending message
Deepak's Chat
Bhejo!

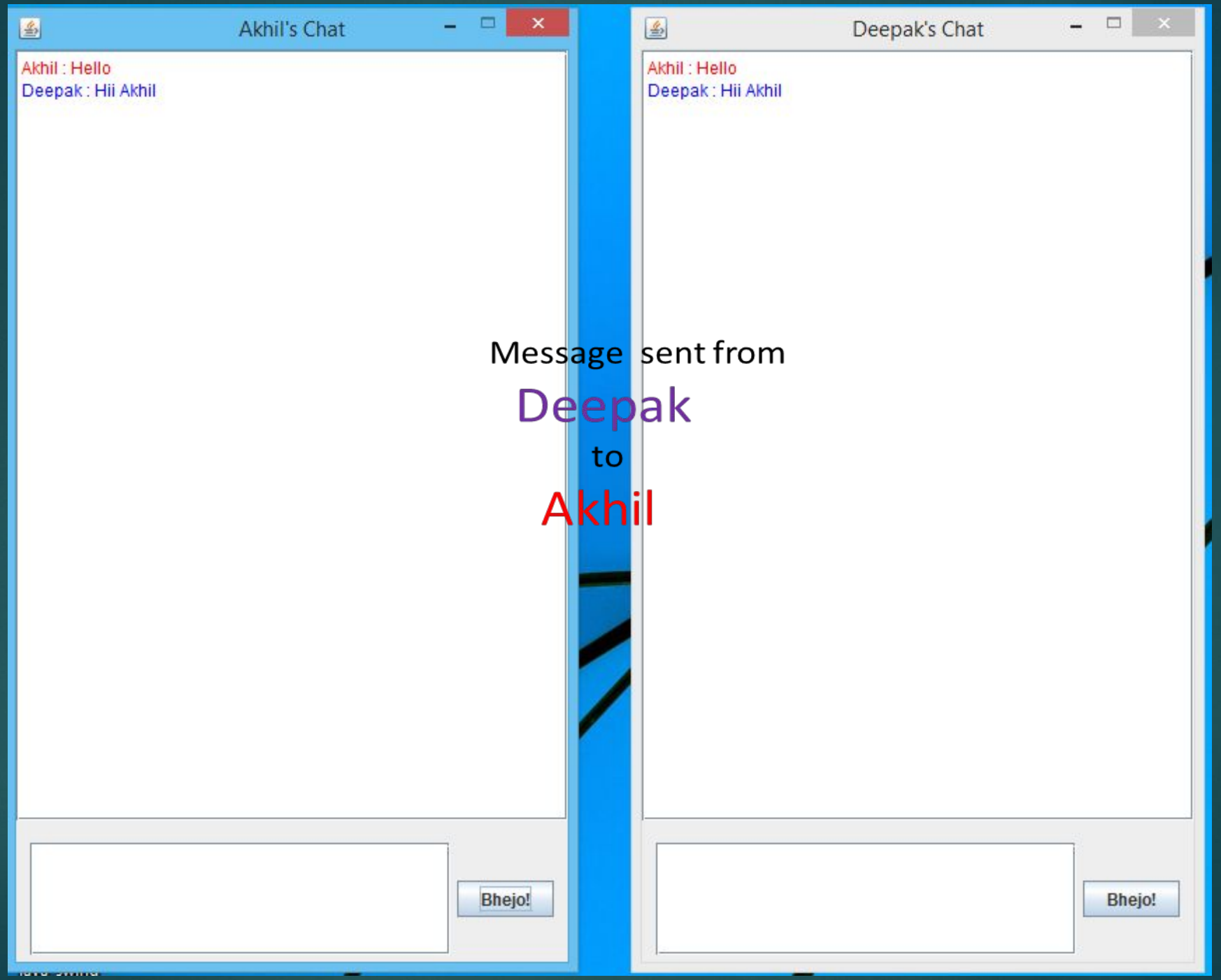


Message sent from

Akhil

to

Deepak



Message sent from

Deepak

to

Akhil